

FILTER DESIGN TEST CASE

Ladder Filter - Band 25 (1900 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Ladder Filter filter targeting Band 25 (1900 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on AlN on SiC substrate with a target center frequency of 1155.7 MHz and bandwidth of 102.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Ladder Filter
- Frequency Band: Band 25 (1900 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: AlN on SiC
- Package: CSP (Chip Scale Package)
- Design Tool: CST Studio Suite 2024
- Design Center: Irvine Design Center

This specification applies to all variants of the Ladder Filter design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1155.7 MHz
Bandwidth (3 dB):	102.9 MHz
Insertion Loss Target:	≤ 3.4 dB
Return Loss Target:	≥ 14.0 dB
Out-of-Band Rejection:	≥ 35.0 dB
Isolation (Duplexer):	≥ 30.0 dB
Q Factor:	1995
Temperature Coefficient:	-26.4 ppm/°C
Die Size:	2.7 x 0.8 mm
Wafer Size:	8-inch
Number of Resonators:	10
Electrode Material:	Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the Ladder Filter template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 1155.7 MHz, BW 102.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on SiC).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3467 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.4 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 154 MHz offset >= 35.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 3.8 MHz
- Power handling shall be >= +34 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-09-18
Devices Tested: 62
Insertion Loss (meas): 3.14 dB
Return Loss (meas): 16.4 dB
Rejection (meas): 38.0 dB
Isolation (meas): 35.3 dB
Q Factor (meas): 1995
Group Delay Variation: 7.7 ns
Power Handling: +36 dBm
Die Yield: 94.9%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Ladder Filter design demonstrated acceptable performance for Band 25 (1900 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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